

THE UNIVERSITY OF THE GAMBIA



DATABASE SYSTEMS II PROJECT REQUIREMENTS (1ST DECEMBER, 2025)

1. Project Overview and Objectives

Objective:

Develop a fully functional database system for a selected real-world application. This project assesses your skills in database design, SQL implementation, data manipulation, optimization, and documentation.

You will also visualize your final database output using a simple **dashboard** based on the data you inserted, **no backend is required**.

This project must be completed in groups of *three (3)* students. Each student must clearly state their individual contributions in the report under a section titled “**Team Roles & Contributions.**”

Sample Project Topics (not limited to):

- School Management System
- Hospital Management System
- Inventory / Asset Management System
- Hotel or Booking Management System
- Any other relevant system of your choice

2. Project Requirements

A. Database Design

You are required to:

- Design a complete **Entity-Relationship Diagram (ERD)** for your system

- Include **at least eight (8) interconnected tables**
- Ensure proper **normalization principles**
- Apply appropriate constraints

B. Data Operations

Using MySQL, you must:

- Insert **at least 10 sample records per table**
- Implement full **CRUD operations** (Create, Read, Update, Delete)
- Write and execute **at least 10 SQL queries**, including:
 - **At least 3 JOIN queries**
 - **At least 2 aggregate queries**
 - **At least 1 ORDER BY or LIMIT query**
 - **At least 1 subquery** (nested SELECT)
- Create and apply **indexing** on at least one table and briefly explain its effect
- Implement **at least one transaction** involving multiple operations
 - Demonstrate both **COMMIT** and **ROLLBACK**
 - Example: inserting or updating multiple related tables in one transaction

C. Dashboard & Data Visualization (15 Marks)

You must create a **simple dashboard** that visualizes your database content. It should include:

- Key metrics (e.g., totals, counts, summaries)
- At least **two charts** (bar, pie, line, etc.)
- At least **one summary table**
- A clear and meaningful representation of the data you inserted into MySQL

Important:

- **No backend is required.**
- You are free to use **any method** to generate the dashboard.
- The dashboard must be based solely on the data you created in MySQL.

D. Tools and Technologies

Use the following:

- **Database System:** MySQL
- **Development Environment:** MySQL Workbench
- **ERD Design Tool:** Any tool of your choice (e.g., Lucidchart, Draw.io, Visio, etc.)

E. Project Report

Submit a well-structured **PDF report** containing the following:

Report Structure

Section	Description
Title Page	Project title, your name, ID, course code, date
Introduction	Overview of your chosen system and objectives
Requirements	Scope and features
ER Diagram	Screenshot + brief explanation
Database Design Description	Explanation of tables, attributes, constraints, and relationships
Query Section	Explanation + screenshots of CRUD and SELECT queries
Indexing	Table where indexing was applied + its impact
Dashboard	Screenshots and explanation of visuals
Conclusion	What you learned, challenges faced, and recommendations

Ensure the report includes **screenshots, diagrams, and code snippets** where necessary.

3. Submission Details

Deliverables:

Submit the following on **Google Classroom** by **11th January, 2026**:

- **Project Report (PDF)**
- **SQL Script File (.sql)** containing:
 - Database creation, Table creation, Sample data, and all implemented queries

Late submissions will incur penalties.

4. Presentation

Your presentation must demonstrate:

- Your ERD and database structure
- Execution of key queries
- Explanation of design decisions and assumptions
- Demonstration of your dashboard
- Understanding of indexing and optimization

5. Grading Rubric Summary

Component	Marks
Database Design	10
Data Operations & Queries	25
Dashboard Visualization	20
Project Report	20
Submission Completeness	10
Presentation	15
TOTAL	100

Plagiarism Policy

Any plagiarism detected will incur a **50% deduction of the total marks**.

ALL THE BEST